SM2501/FSG3084

Measuring and analyzing motion

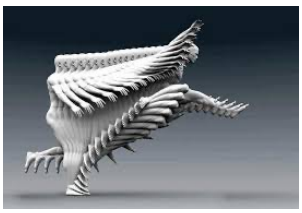
Lecture 3



E. J. Marey 1830-1904



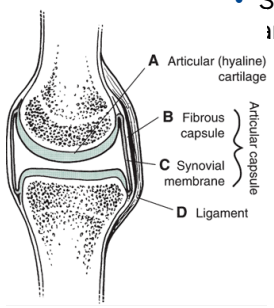
1962 image of dancer
dancer Patricia Mcbride
-Gjon Mili 1904-1984



Peter Jansen
1956-

Review

- Bone tissue and properties
- Joints
- Soft tissue composition and properties

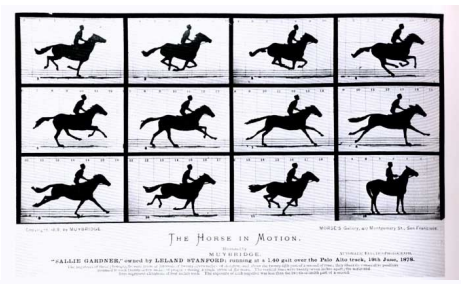


Hamill & Knutzen 2009



We will come back to muscle....

Motion Capture and Analysis

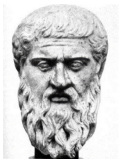


EADWEARD MUYBRIDGE. *Galloping Horse*. 1878. Albumen print. George Eastman House, Rochester, N.Y.

Objectives

- Describe the historical development of motion analysis
- Compare advantages/disadvantages of different motion capture technologies
- Explain the workflow from marker data to kinematic outputs
- Identify coordinate systems and reference frames commonly used in movement analysis
- Form basis to interpret motion capture data

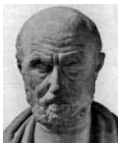
3



Plato (428-348 BC)

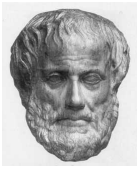
the head, being the most divine part of us … the gods, when they put together the body, gave all the other members to be servants

Hippocrates (460-370 BC)



- *When, then, a dislocation has not been reduced, but has been misunderstood or neglected, … the weight of the body is mostly supported on the sound leg*
- *One, then, is forced in walking to turn the (sound) leg inward, and not outward, for thus the sound leg best supports its own half of the body, and also that of the injured side*

4



Aristotle (384-322 BC)

- *If a man were to walk parallel to a wall in sunshine, the line described (by the shadow of his head) would be not straight but zigzag, becoming lower as he bends, and higher when he stands and lifts himself up*
- *all creatures naturally have the power of changing position by the use of limbs*

5



Leonardo da Vinci (1452-1519)

- Studied center of gravity and balance
- Retrieved corpses from cemeteries in the middle of the night and identified muscles and nerves.
- He suggested that cords be attached to a skeleton at the points of origin and insertion of the muscles, to describe interaction of muscles.



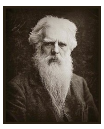


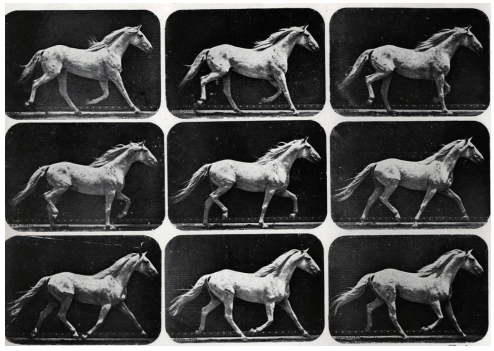
Giovani Alfonso Borelli (1608-1679)

- Studied center of mass with platform and knife edge

6

Eadward Muybridge (1830-1904)







Tasked by Leland Stanford to solve a debate on whether a galloping horse ever had all four feet off the ground (answer: yes)

Numerous cameras in a line, shutters triggered by thread from horse

This is called the first film

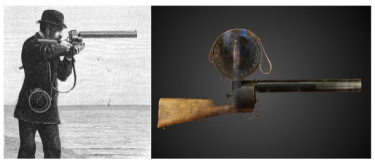
7

Étienne-Jules Marey (1830-1904)



All movement is the product of two factors: time and space; to know the movement of a body is to know the series of positions that it has occupied in space in a series of successive moments

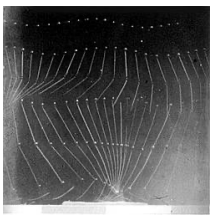
1882: chronophotographic gun :
12 fps on same image













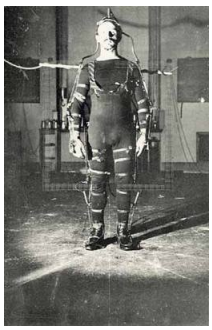
Marcel Duchamp,
1912
Nude descending a staircase

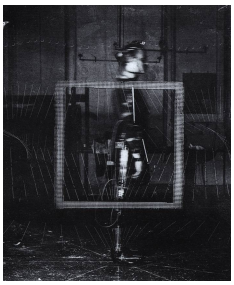


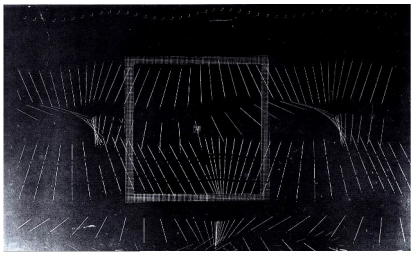
8

Braune and Fischer (1895)

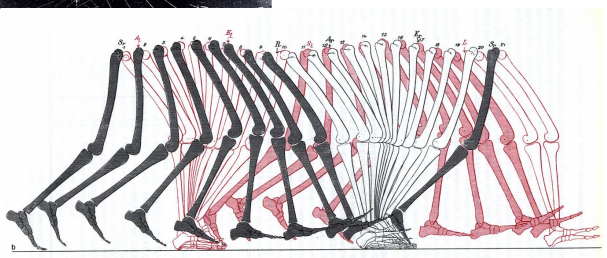
First 3D analysis of human gait, strobes
~8 hours to prepare
Several synched cameras from different angles







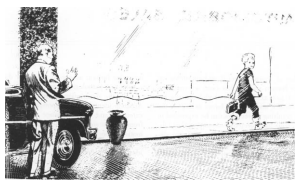
"The Human Gait" (1895-1904)



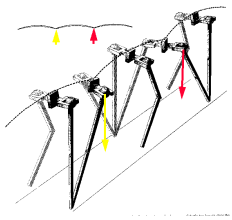
9

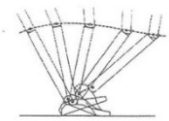
Verne Thompson Inman (1905-1980)
Howard Davis Eberhart (1906-1993)
John B. De C. M. Saunders (1903-1991)

'Determinants of Gait', **J Bone Joint Surg 1953**







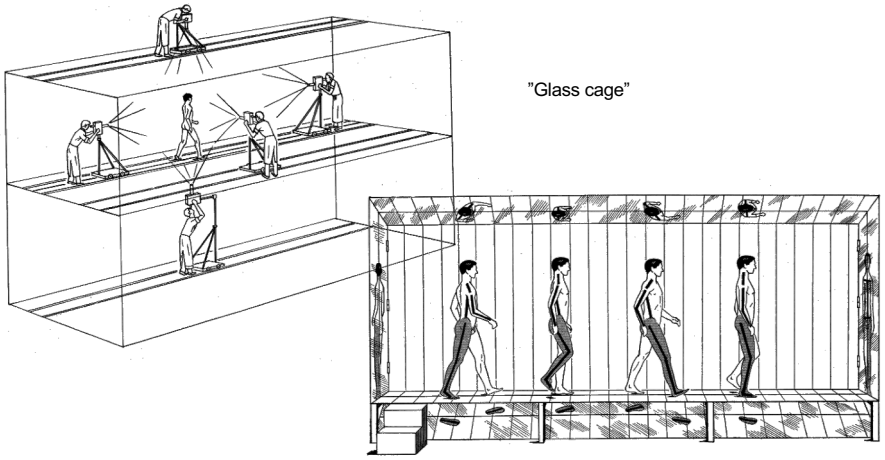


- Focus on center of mass oscillations
- Smaller and smooth motions are more energy efficient than inverted pendulum
- Detailed 6 'determinants' that make COM trajectory smooth

10

Charles, Robert, Jean & Pierre Ducroquet.

"Walking and limping" 1965



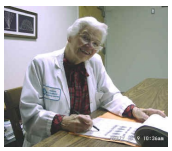
11

The recents and currents

James R. Gage (193?-)

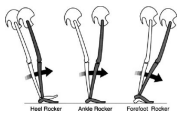


- "The treatment of gait problems in cerebral palsy"
- "Gait analysis in cerebral palsy"
- "The identification and treatment of gait problems in cerebral palsy"



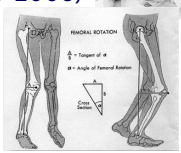
Jacquelin Perry (1918-2013)

"Gait Analysis: Normal and Pathological Function"



David Sutherland (1923-2006)

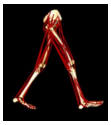




Scott Delp (1965? -)

OpenSIM, "Biomechanics of Movement"

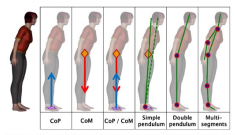




David Winter (1930-2012)

"Biomechanics and Motor Control"





And many others.....

12

Biomechanics and human movement today



National societies associated with
International Society of Biomechanics (ISB)

Pubmed search: 'Biomechanics' and 'Human Movement': over 7500 entries. Around half in last 10 years.

Practically all universities and larger hospitals

ISB conference – human movement biomechanics ~1600 participants 2025

Several motion analysis labs at KTH (MoveAbility, media technology, marine and aerospace engineering)

13

Why do we measure motion?

Research applications:

- Understanding fundamental movement mechanics and control
- Injury mechanism identification and prevention strategies
- Performance optimization in sports and rehabilitation
- Prosthetic and orthotic device design and evaluation
- Workplace ergonomics and injury prevention

Clinical applications:

- Gait analysis for surgical planning (e.g., cerebral palsy, stroke)
- Rehabilitation progress monitoring, pre- and post-surgical outcome evaluation
- Fall risk assessment in elderly populations
- Quantifying movement dysfunction in neurological disorders

Industrial applications

- Animation and gaming (motion capture for realistic character movement)
- Human-robot interaction design
- Vehicle and workspace design (ergonomics)
- Athletic training and technique analysis

14

Methods to capture motion

15

Videos + 2D pose estimation (~10-50 mm accuracy)

Technologies:

- OpenPose, MediaPipe, DeepLabCut, AlphaPose, etc.
- Convolutional neural networks trained on large datasets



+ Good for large scale studies, multiple subjects
+ No specialized equipment, cheap, easy
+ Many open-source options
+ Can be real-time
+ Can analyze archived video footage
- Depth ambiguity
- Sensitive to clothing, lighting, camera angle
- Trained primarily on specific populations (bias concerns)
- Questionable reliability, validity and generalizability

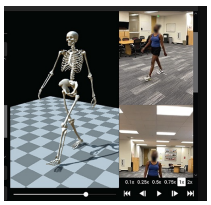
16

Markerless 3D, multiple calibrated cameras (~10-30 mm accuracy)

Technologies:

- Theia3D, OpenCap, FreeMoCap
- Triangulation from multiple (~2-8) synchronized cameras
- Can use standard RGB cameras (smartphone, GoPro, etc)





+/- More accurate than 2D, less accurate than marker-based
Joint angle accuracy: ±3-5° (vs. ±0.5-1° for marker-based)
+/- May or may not be pricy
+ Can be real-time or near real-time
+ Possibility to analyze archived video footage
- Training algorithms may introduce bias
- Lab and calibrated cameras/devices still required
- Occlusion problems
- Generalizability across body types and clothing

17

Depth camera systems

(Joint angle accuracy ~5-15°, Depth accuracy: 1-5 mm at 1m, degrades with distance)

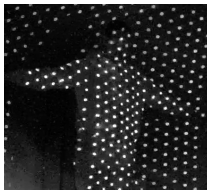
Technologies:

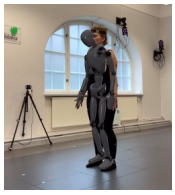
- Azure Kinect, Intel RealSense, iPhone LiDAR + ARKit
- Time-of-flight or structured light sensors

How they work:

- Emit infrared light, measure return time
- Create depth map + RGB image
- Combined with skeletal tracking algorithms

+ Pre-installed on many smart devices + standalone cameras
+ Real-time kinematics estimation
+ Very easy to use
- Training algorithms may introduce bias
- Own training algorithms can increase accuracy
- Unclear reliability, validity and generalizability
+ Some indication that for future sensor fusion

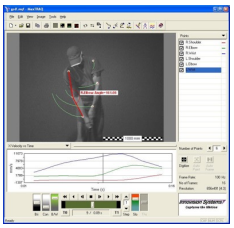


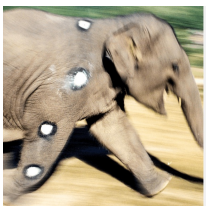


18

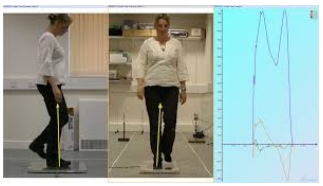
Videos with or without force sensors

Video-based 2D motion digitization
+ Cheap
+ Little or no instrumentation
+ Visually explanatory
- 2D, specialized software
- Low accuracy
- No kinetics





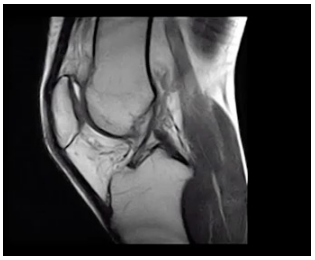
Video-vectors
+ Relatively cheap
+ Rel. little instrumentation
+ Realtime
- Not entirely 3D
- Can be misleading quantitively

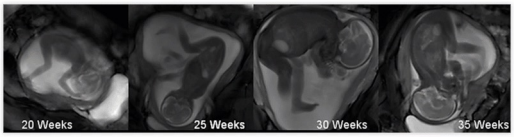


19

Non-ionizing imaging

Dynamic MRI
+ Soft tissues easily visible
+ Can be in reconstructed in 3D (4D)
+ Can apply markerless technologies
- Usually unloaded position
- Limited motions
- Pricy and time-consuming





Verbruggen et al, 2018

20

Ionizing imaging

X-ray methods (Very accurate <1 mm, 1.5-2°)
- Radiostereometric analysis (RSA) or biplanar X-ray
  - + Low radiation
  - - Static
- Biplanar Flouroscopy
  - + Dynamic x-ray
  - + Soft tissues not visible





Dynamic Computed tomography (CT)
- + Bones clearly visible
- + Can be 3D (4D) reconstructed
- + Can apply markerless technologies
- - Usually unloaded position
- - Limited motions
- - Pricy and time-consuming



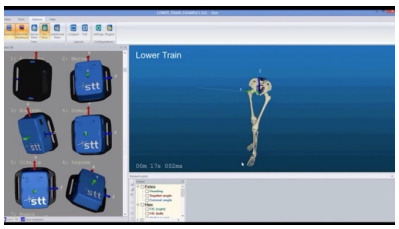


21

Wearable inertial sensors

IMUs: accelerometer (linear accel) + gyroscope (angular veloc) + magnetometer (magnetic field orientation)
- + Wireless, lightweight, portable, no lab required
- + Widely available for large studies (e.g. Fitbit, smartphones)
- + Some can store data for hours
- +/- Measures displacement from reference position
- +/- Can be expensive
- - Subject to drift
- - Subject to soft tissue artifact
- - Joint kinematics accuracy 3°- 10°, plus drift for research-grade





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Cameras + marker-based systems (<1 mm, 0.5-1° accuracy)

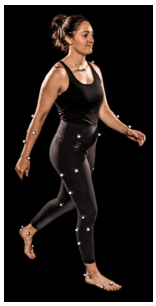
+ Much more accurate than without markers
+ Complex 3D motion
+ High sampling frequency
- Requires lab
- Requires modelling
- Soft tissue artifact
- Relatively expensive

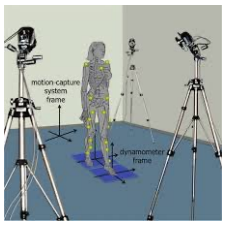




Active markers (cameras are receivers)
+ Large capture volumes
+ Visibility less of problem than with passive
- Usually some wires

Passive markers (cameras are senders and receivers)
+ No cables
+ Relatively quick and unobtrusive
- Markers/subjects must be identified
- Many cameras required for visibility





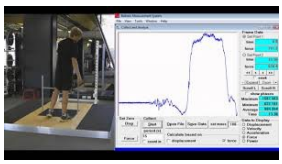
23

Complementary or standalone techniques often combined with movement analysis

24

Force or pressure measurement

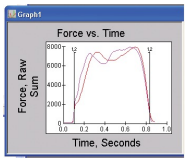
Force plates
- + Very accurate (<1% error)
- + Little instrumentation
- + High sampling frequency, real-time
- - Good quality is expensive

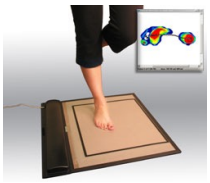


Pressure/Pedabarography
- + Relatively little instrumentation
- + Realtime
- + Illustrative
- - Only 1D pressure (no shear)
- - Requires models to distiguish foot segments
- - Tendency for drift
- - High resolution is expensive





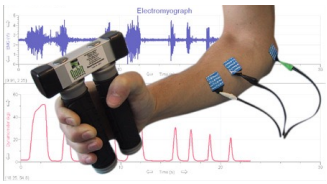


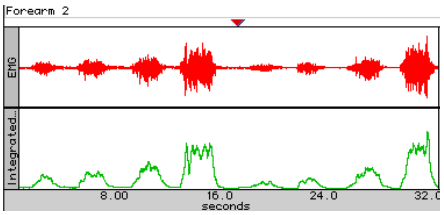


25

Muscle excitation measurement

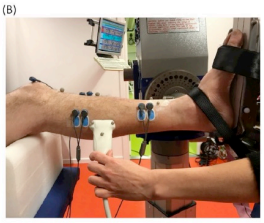
Electromyography (EMG)
- + Little instrumentation
- + Real-time
- + Identifies physiological causality of motion
- +/- Noisy if not placed securely
- - Surface EMG: crosstalk is a problem
- - Muscle activation related to but ≠ muscle force

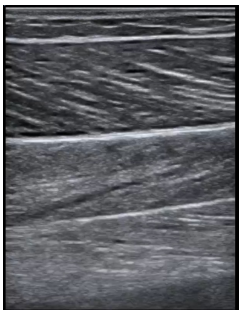




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Ultrasound





R Wang

+ Can view/measure tissues
+ If linked to motion capture, makes tissue-scale studies possible
- Quantitative reliability not excellent
- 3D but more reliable in superficial than deep muscles

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Metabolic cost measurement

Spiroergonometry



Measures O_2 and CO_2 in inhalations and exhilations, estimates metabolic consumption and cost

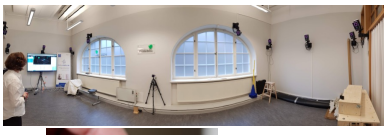
+ Usually measures pulse and GPS
+ Portable, lightweight
+ High reliability and validity
- Computing metabolic cost requires achieving steady state (3-5 minutes)
- Breath-by-analysis: can be less than 1 sample/second
- Influenced by many physiological factors
- Possibly intrusive

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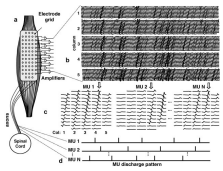
System at Promobilia MoveAbility Lab

Motion analysis (10 Vicon V16, 3 Vero cameras)
4 fixed + *3 portable force plates (AMTI)
*IMUs (Myon aktos-t)
*EMG (Myon aktos-nano)
*Ultrasound (Philips)
*Pressure insoles (pedar, Moticon)
*Depth cameras/LiDAR
Treadmill instrumented with force plates (AMTI)
*High-density EMG 400 + 64 channels (OTB)
*Metabolic cost measurement (Cortex Metamax)

* Modular/portable

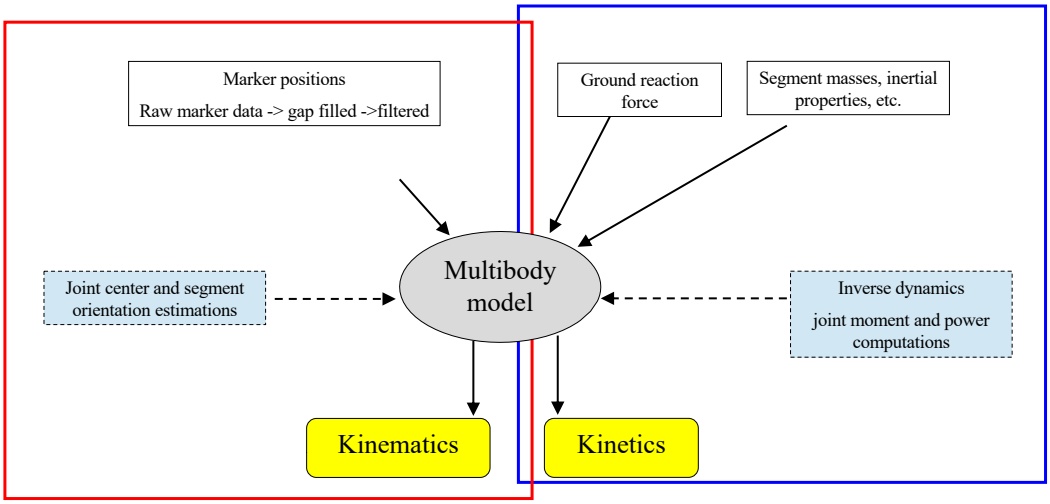





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Experimental motion analysis

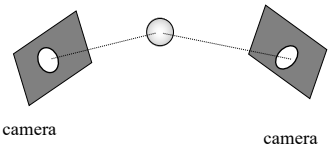


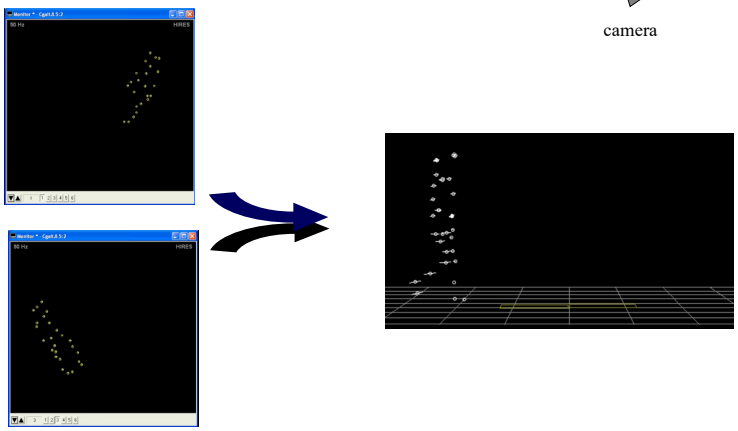
With optional further information from EMGs, video, pressure distribution, metabolic cost measurements, etc.

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Kinematics from marker-based systems

3D position computed from ≥2 calibrated cameras via triangulation





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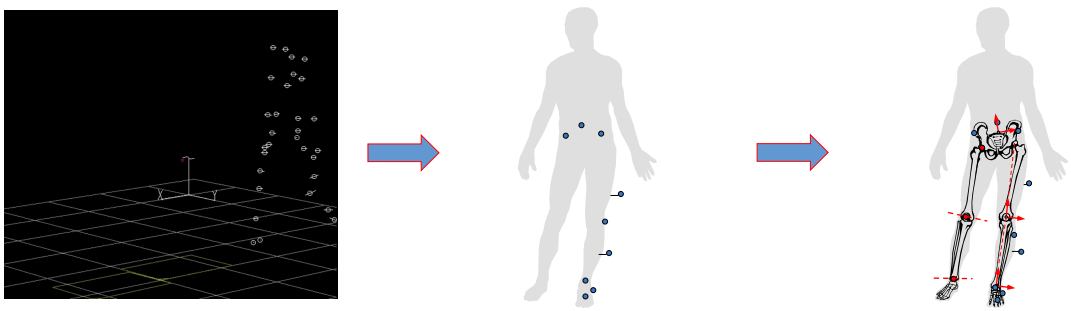
Coordinate systems, multibody model, reference frames

3D marker position in **global** coordinate system

Markers identify
- Anatomical landmarks
- Segment axes/orientation

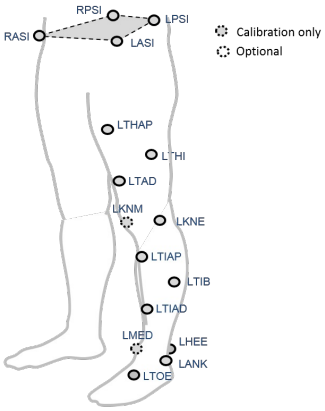
Virtual points
Segments: min 3 markers for 3D
Joints: as per model + constraints

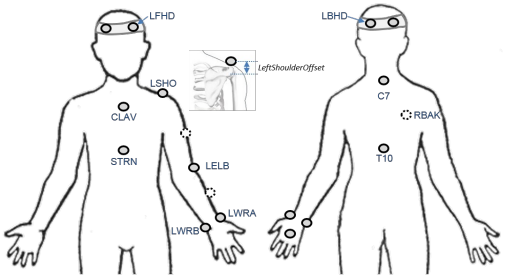
Anatomical coordinate frames: fixed to body segments, aligned with anatomical directions



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Marker set ↔ Model CGM 2.3 model



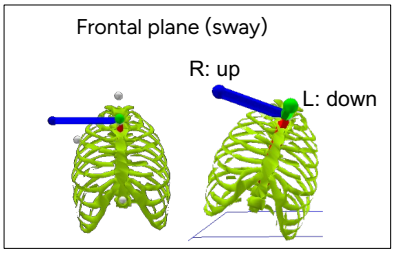


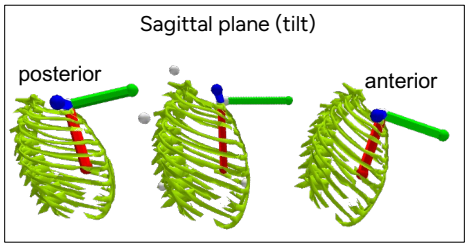
https://pycgm2.netlify.app/
Leboeuf et al, Gait Posture 2019

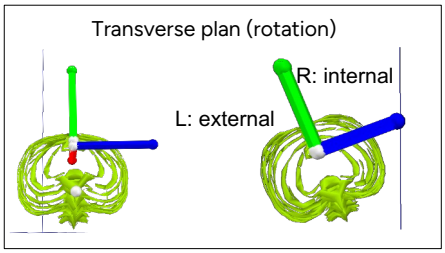
Hips, knees: ball-and-socket (spherical) joint. 3D rotation, no translation. OpenSim: knee might be modelled as hinge.
Ankle: depends, often 2D (sagittal, transverse). In OpenSim: talocrural joint is hinge unless subtalar joint is unlocked
Spine/back: depends. Sometimes not actual joint, sometimes entire spine/trunk modelled as rigid

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Trunk
Defined in global reference frame



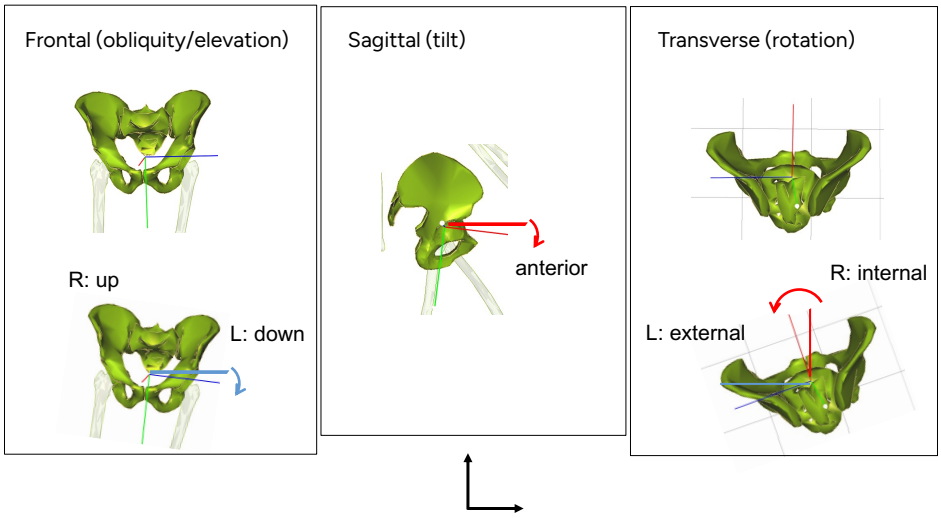




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Pelvis

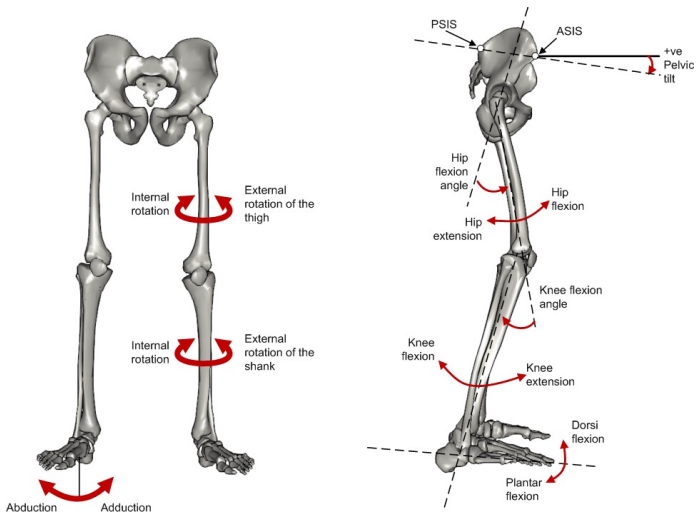
Defined in global reference frame



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Lower extremity joints

Relative angles, defined in proximal segment's reference frame



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Reminder: tomorrow

Compulsory to attend one session.

Promobilia MoveAbility Lab
Osquars Backe 18, 4th floor. Just across from library entrance

Come 5 min before scheduled time

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